

Developing Explainable AI models for DDoS Network Intrusion Detection Systems

Undergraduate Thesis

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Background

- Integrity and Availability are two key pillars of cybersecurity
- Zero Trust Network Systems require application, user, device, and data level transparency
- Machine Learning Classifiers tasked with Network Security are by default black box systems
- An explainable model is required for Zero Trust Systems; a white box model

Distributed Denial of Service [DDoS]

- A malicious user disrupts a target server's ability to process requests
- Network intrusion detection systems monitor network traffic for malicious activities at the firewall application level

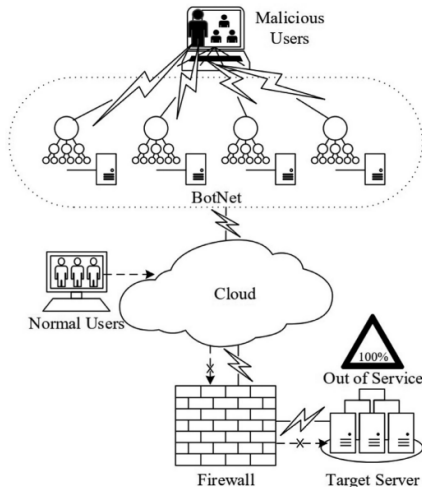


Figure: DDoS Process [1]

Network Intrusion Detection Systems [IDS]

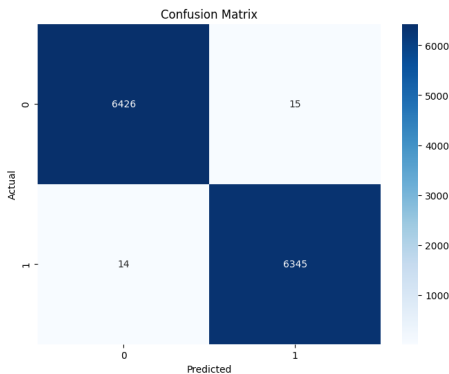


Figure: Anomaly Detection Confusion Matrix (benign, malicious)

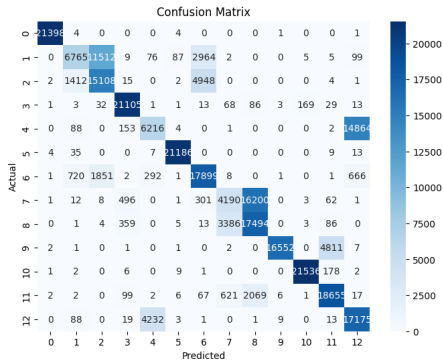


Figure: Signature Detection Confusion Matrix (benign, subsequent types of attacks)

- CICDDoS2019 Dataset
- CICFlowMeter-V3
- Packet flow data from source to destination

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13 -----
14 List of extracted features and descriptions:
15 Feature Name          Description
16 Flow duration          Duration of the flow in Microsecond
17 total Fwd Packet       Total packets in the forward direction
18 total Bwd packets      Total packets in the backward direction
19 total Length of Fwd Packet Total size of packet in forward direction
20 total Length of Bwd Packet Total size of packet in backward direction
21 Fwd Packet Length Min  Minimum size of packet in forward direction
22 Fwd Packet Length Max  Maximum size of packet in forward direction
23 Fwd Packet Length Mean Mean size of packet in forward direction
24 Fwd Packet Length Std  Standard deviation size of packet in forward direction
25 Bwd Packet Length Min  Minimum size of packet in backward direction
26 Bwd Packet Length Max  Maximum size of packet in backward direction
27 Bwd Packet Length Mean Mean size of packet in backward direction
28 Bwd Packet Length Std  Standard deviation size of packet in backward direction
29 Flow Bytes/s           Number of flow bytes per second
30 Flow Packets/s         Number of flow packets per second
31 Flow IAT Mean          Mean time between two packets sent in the flow
32 Flow IAT Std           Standard deviation time between two packets sent in the flow
33 Flow IAT Max           Maximum time between two packets sent in the flow
34 Flow IAT Min           Minimum time between two packets sent in the flow
35 Fwd IAT Min            Minimum time between two packets sent in the forward direction
36 Fwd IAT Max            Maximum time between two packets sent in the forward direction
37 Fwd IAT Mean           Mean time between two packets sent in the forward direction
  
```

Figure: Sample of CICDDoS2019 Features [2]

- Exclude WebDDoS sample (not enough data)
- Remove entries with non-numeric feature values
- Remove features with only 0s (resultant 69 features)
- Perform scalar normalization on each feature
- Reservoir sample 100,000 data points from each class

Attack Taxonomy

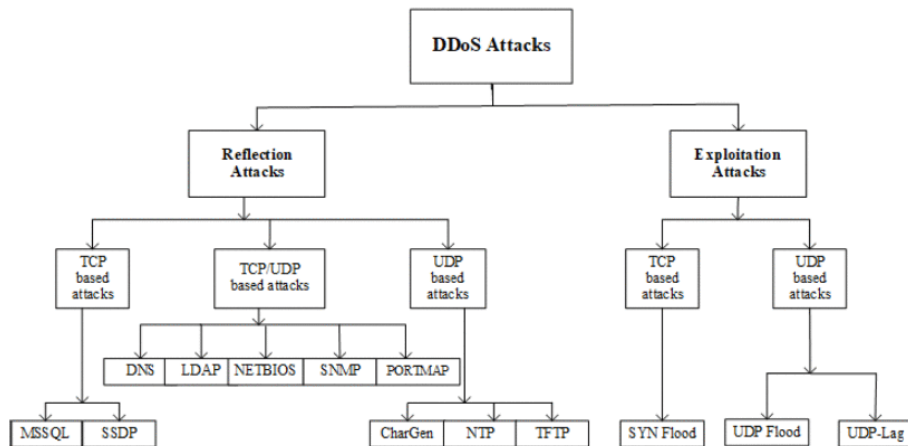


Figure: CICDDoS attack taxonomy defined by Sharafaldin et al.[2]

Reflection Attacks

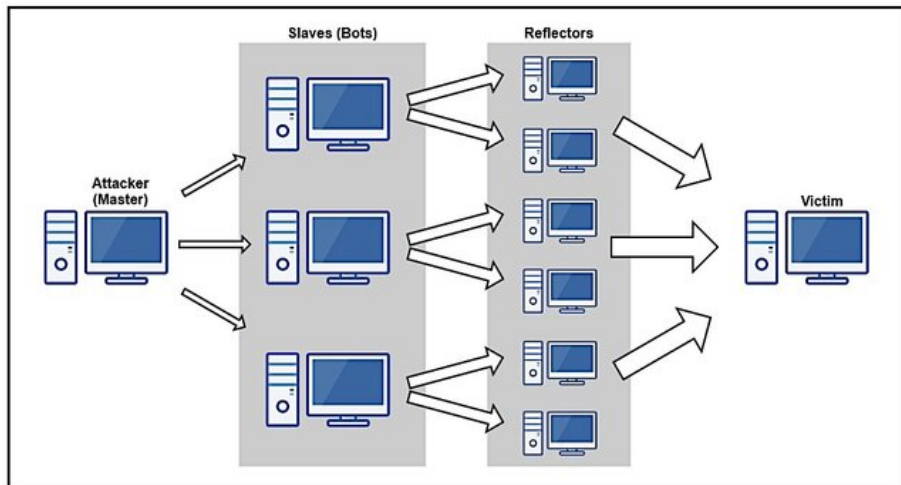


Figure: Amplification reflection attack diagram [3]

Exploitation Attacks

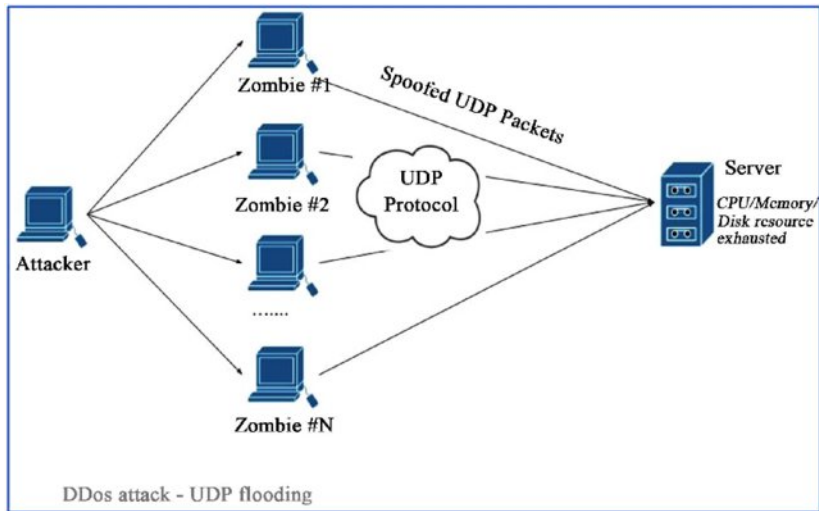


Figure: Exploitation flood attack diagram [4]

Objectives & Constraints

- Objective: Models must achieve a reasonable performance based on prior works (99% Accuracy for anomaly detection)
- Objective: Develop a method for which models' predictions can be understood without additional post hoc explanation. Metric of choice: SHAP Graph Edit Distance
- Constraint: Computational Resources, constraint criteria is model must be sufficiently trainable with on-hand resources (AMD Ryzen 7 3800XT 8-Core CPU, NVIDIA GeForce RTX 3070 8Gb VRAM GPU)
- Constraint: Complexity, the model must be reasonably compact to minimize overhead of explainability tools, and also to minimize prediction latency when observing network traffic through CICFlowMeter-V3. Metric of choice: number of model layers and trainable parameters.

Models

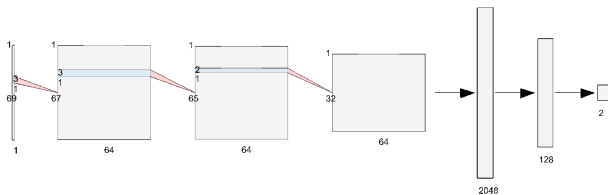


Figure: Anomaly Detection Model Architecture

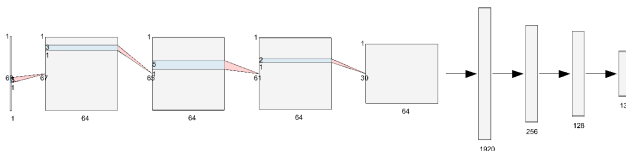


Figure: Signature Detection Model Architecture

Shapley Values

Shapley value definition [5]:

$$\phi_j(val) = \sum_{S \subseteq \{1, \dots, p\} \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} (val(S \cup \{j\}) - val(S)) \quad (1)$$

$$val_x(S) = \int \hat{f}(x_1, \dots, x_p) d\mathbb{P}_{x \notin S} - E_x(\hat{f}(X)) \quad (2)$$

$$\sum_{j=1}^p \phi_j = \hat{f}(x) - E_X(\hat{f}(X)) \quad (3)$$

Anomaly Detection Knowledge Graphs (KG)

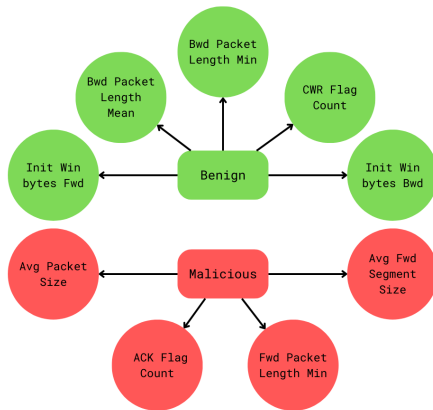


Figure: Binary Classification Knowledge Graph

Signature Detection Knowledge Graphs (KG)

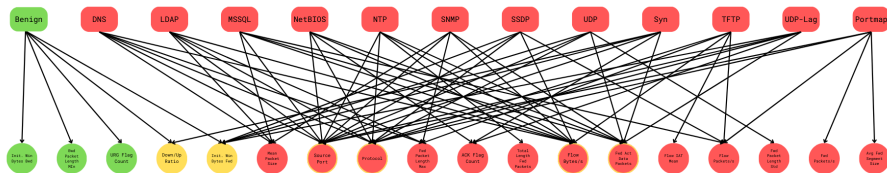


Figure: Topk Knowledge Graph, $k=5$

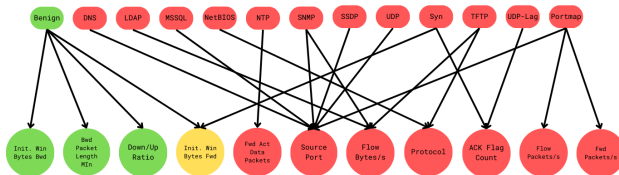


Figure: Pruned Knowledge Graph

Algorithm Applies to the loss function the polarity difference between sampled SHAP values, and the ground truth KG

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procedure  $Loss_{SHAP}(L, \hat{y}, \phi_{\hat{y}}, KG, h)$   
  while  $l < num\_labels$  do  
     $\phi_l = \sum \phi_{\hat{y}}, \text{ where } \hat{y} = l$   
  end while  
   $\phi = [\phi_0, \phi_1, \dots, \phi_l]$   
   $\beta = \sum (-KG \cdot \phi)^+$      $\triangleright$  Keep only KG misaligned SHAP attribution  
  return  $L * (h\beta + 1)$   
end procedure
```

Modified definition from neuro-symbolic learning [6]

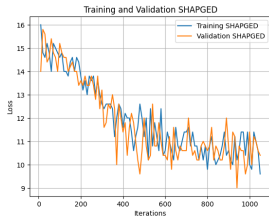
SHAP Graph Edit Distance (GED)

Algorithm Finds the edge difference between a ground truth KG and a SHAP generated KG

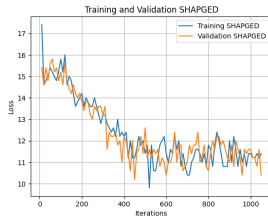
```
procedure  $GED_{SHAP}(L, y, \hat{y}, \phi_y, KG, t)$   
  while  $l < num\_labels$  do  
     $\phi_l = \sum \phi_{\hat{y}}$ , where  $\hat{y} = l = y$   
     $\phi_l = \frac{\phi_l}{|\max(\phi_l)|}$  ▷ Normalize feature attributions per class  
  end while  
   $\phi = [\phi_0, \phi_1, \dots, \phi_l]$   
   $\phi = -1$ , where  $\phi < t$  ▷ Mask Shap values  
   $\phi = 1$ , where  $\phi \neq -1$   
  return  $\sum 1$ , where  $KG \neq \phi$  ▷ simplified GED  
end procedure
```

Modified definition from neuro-symbolic learning [6]

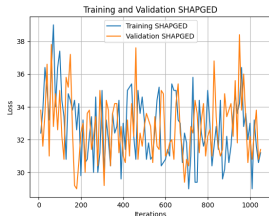
SHAP Graph Edit Distance



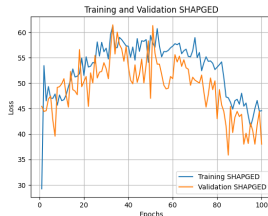
(a) Anm. Det. Baseline



(b) Anm. Det. $L_5(h=1)$

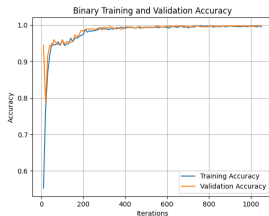


(c) Sig. Det. Baseline

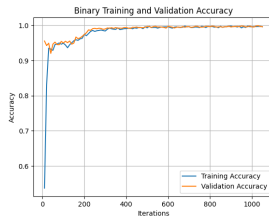


(d) Sig. Det. $L_5(h=1)$

Performance



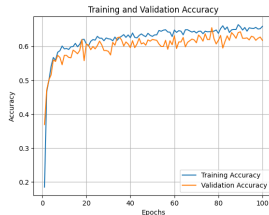
(a) Anm. Det. Baseline



(b) Anm. Det. $L_5(h=1)$

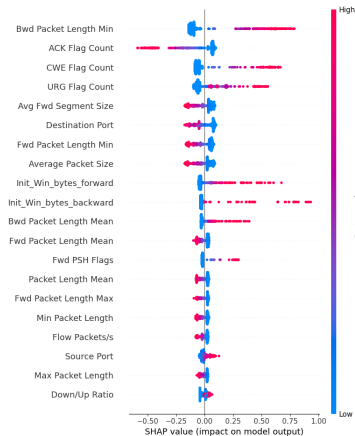


(c) Sig. Det. Baseline

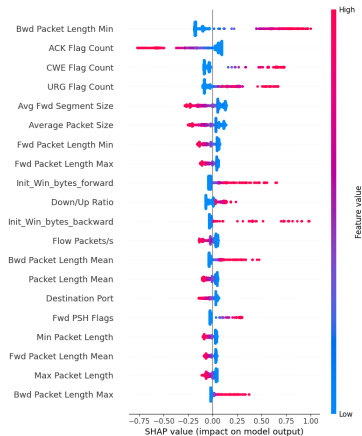


(d) Sig. Det. $L_5(h=1)$

Binary Experiment SHAP Values

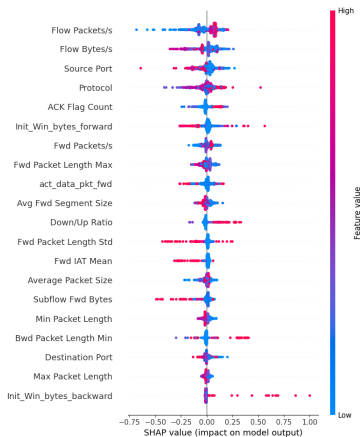


(a) Anm. Det. Baseline
(Benign Prediction)

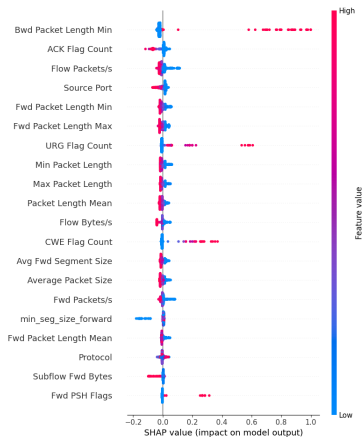


(b) Anm. Det. $L_5(h=1)$
(Benign Prediction)

Multiclass Experiment SHAP Values

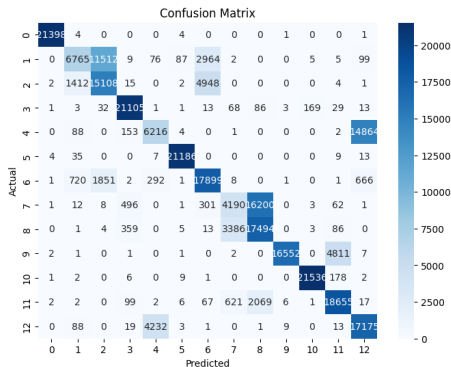


(a) Sig. Det. Baseline
(Benign Prediction)

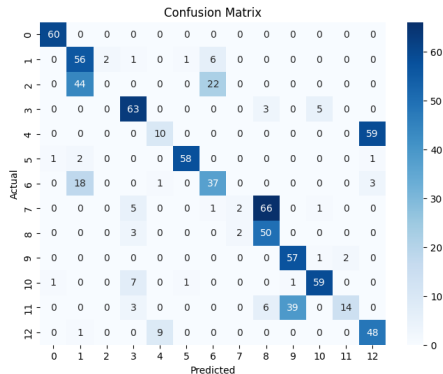


(b) Sig. Det. $L_S(h = 1)$
(Benign Prediction)

Signature Detection Predictions



(a) Sig. Det. Baseline



(b) Sig. Det. $L_5(h=1)$

Primary Experiment Takeaways

- Affirmed SHAP Loss model matches baseline performance and explainability when expected important features closely resembles baseline model SHAP Attributions.
- Improved clarity of local SHAP for signature detection via SHAP Loss and SHAP GED, although the method trades performance for explainability.

Next Steps

- Test and tune h parameter in SHAP Loss to try and balance performance with explainability
- Acquire and test more knowledge graphs from subject experts
- Split ID number based data features via binning and retest method (prior work shows that similar methods function better when classes have features unique to them)

Questions?

References

-  D. Akgun, S. Hizal, and U. Cavusoglu, “A new ddos attacks intrusion detection model based on deep learning for cybersecurity,” *Computers & Security*, vol. 118, p. 102748, 2022.
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